

ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

THEORY OF COMPUTATION

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 3rd year, semester 2 final examination: finite automata, regular expressions, non-regular languages, context-free grammars and pushdown automata, Turing machines, decidability and complexity.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (theory of computation: automata, computability and complexity). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. In state diagrams, an arrow from nowhere marks the start state and a double circle marks an accept state.
3. ε stands for the empty string and ␣ for the blank symbol of a Turing machine. In regular expressions, ∪ means "or".
4. Draw state diagrams neatly and label every transition.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which kind of machine recognises exactly the regular languages?

A Pushdown automaton
B Finite automaton
C Turing machine
D Linear bounded automaton

1.2 Which of these languages is NOT regular?

A $\{a^n b^n \mid n \geq 0\}$
B strings over {a, b} with an even number of a's
C a*b*
D strings over {a, b} that end in ab

1.3 Context-free languages are NOT closed under …

A union.
B concatenation.
C intersection.
D star.

1.4 The subset construction turns an NFA with 4 states into a DFA with at most …

A 4 states.
B 8 states.
C 16 states.
D 32 states.

1.5 Which regular expression describes all strings over {a, b} that contain ab?

A $(a \cup b)^*ab(a \cup b)^*$
B a^*b^*
C ab^*
D $(ab)^*$

1.6 The language $A_{TM} = \{\langle M, w\rangle \mid M \text{ is a Turing machine that accepts } w\}$ is …

A decidable.
B Turing-recognisable but not decidable.
C not Turing-recognisable.
D regular.

1.7 A pushdown automaton is a nondeterministic finite automaton with an extra …

A queue.
B second tape.
C stack.
D counter.

1.8 Which problem is known to be in P?

A PATH: is there a path from s to t in a directed graph?
B SAT: is a Boolean formula satisfiable?
C HAMPATH: does a graph have a Hamiltonian path?
D 3-colouring of a graph

1.9 If $A \leq_P B$ (A reduces to B in polynomial time) and B is in P, then …

A A is in P.
B A is NP-complete.
C B is NP-hard.
D nothing can be said about A.

1.10 The Church–Turing thesis states that …

A every algorithm can be carried out by a Turing machine.
B P = NP.
C every Turing machine halts.
D Turing machines are faster than computers.

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SECTION B

QUESTION 2: DETERMINISTIC FINITE AUTOMATA

FIGURE 1 shows the DFA M1 over the alphabet $\Sigma = \{0, 1\}$.

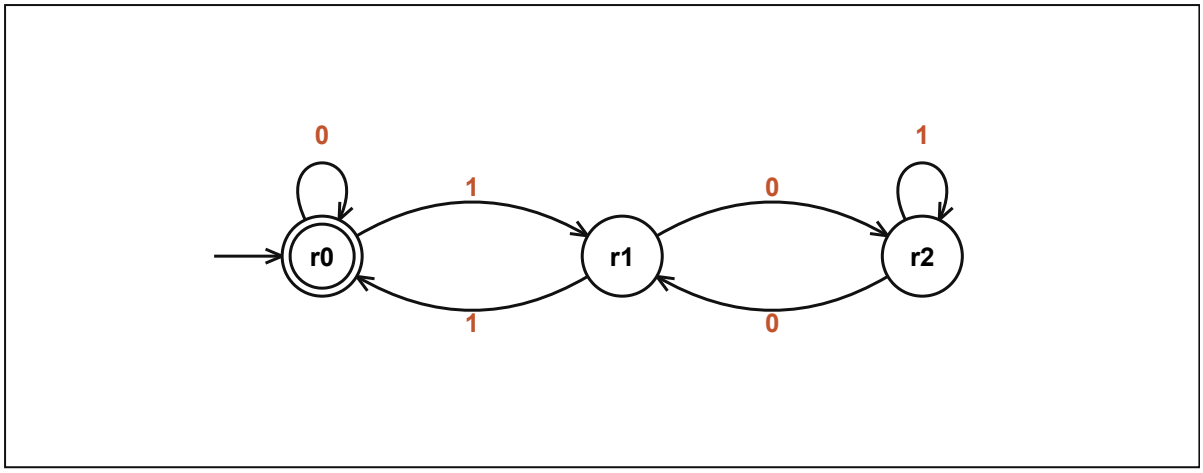


FIGURE 1: DFA M1

2.1 Answer the questions about M1.

2.1.1 Give the formal definition of M1 as a 5-tuple $(Q, \Sigma, \delta, q_0, F)$. Give δ as a table. (4)

2.1.2 Give the sequence of states M1 goes through on the inputs 1001 and 1011. Is each input accepted or rejected? (3)

2.1.3 Describe in words the language that M1 recognises. (Hint: read each input as a binary number.) (2)

2.2 Draw the state diagram of a DFA over $\Sigma = \{a, b\}$ that recognises each language:

2.2.1 $\{w \mid w \text{ contains the substring aba}\}$ (4)

2.2.2 $\{w \mid w \text{ has an even number of a's and an odd number of b's}\}$ (4)

2.3 Explain how to turn a DFA for a language L into a DFA for its complement. What does this show about regular languages? (3)

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QUESTION 3: NONDETERMINISM AND REGULAR EXPRESSIONS

FIGURE 2 shows the NFA N over $\Sigma = \{0, 1\}$.

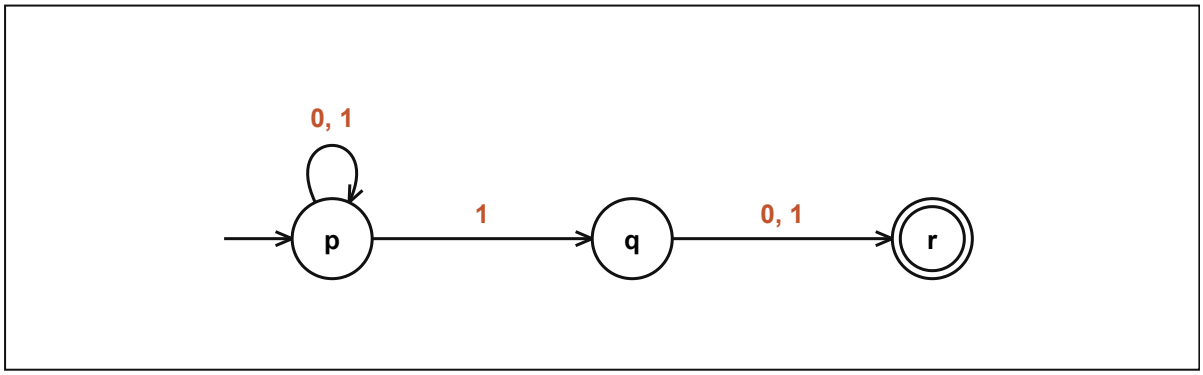


FIGURE 2: NFA N

3.1 Answer the questions about N.

3.1.1 Is the string 0110 accepted by N? Give an accepting path, or explain why there is none. (2)

3.1.2 Use the subset construction to convert N into an equivalent DFA. Show only the states that can be reached from the start state, give the transition table, and mark the start and accept states. (6)

3.2 Give a regular expression for each language over the given alphabet:

3.2.1 Strings over {a, b} that begin with a and end with b. (2)

3.2.2 Strings over {0, 1} that do not contain two 1s next to each other. (2)

3.3 Every NFA can be converted to a DFA. Give TWO reasons why NFAs are still useful. Also state the price paid when an NFA is converted to a DFA. (3)

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QUESTION 4: NON-REGULAR LANGUAGES

4.1 State the pumping lemma for regular languages. (4)

4.2 Use the pumping lemma to prove that $L = \{0^n1^n \mid n \geq 0\}$ is not regular. (6)

4.3 Use closure properties and the result of 4.2 to prove that $EQ = \{w \in \{0, 1\}^* \mid w$ has the same number of 0s and 1s$\}$ is not regular. (3)

4.4 Is every finite language regular? Explain your answer. (2)

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QUESTION 5: CONTEXT-FREE LANGUAGES

The grammar G has start variable S and the rules $S \to aSb \mid SS \mid \varepsilon$.

5.1 Answer the questions about G.

5.1.1 Give a leftmost derivation of the string aabbab. (3)

5.1.2 Show that G is ambiguous. (3)

5.1.3 Describe L(G) in words. (Hint: think of a as "(" and b as ")".) (1)

5.2 Write a context-free grammar for each language:

5.2.1 $\{a^nb^{2n} \mid n \geq 0\}$ (2)

5.2.2 All palindromes over {0, 1} (strings that read the same forwards and backwards). (2)

5.3 Draw the state diagram of a pushdown automaton that recognises $\{0^n1^n \mid n \geq 0\}$. Write each transition as "input, pop → push". (5)

5.4 Explain why compilers use context-free grammars, and give ONE rule of a programming language that a context-free grammar cannot check. (3)

5.5 Give an example of a language that is not context-free. (1)

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QUESTION 6: TURING MACHINES, DECIDABILITY AND COMPLEXITY

The Turing machine M has the states q0 (start), q1 and q_{acc}, and the transitions below. The input is written on the tape with blanks ⊔ on both sides, and the head starts on the first input symbol.

State	Reads 0	Reads 1	Reads ⊔
q0	write 1, move R, stay in q0	write 0, move R, stay in q0	write ⊔, move L, go to q1
q1	write 0, move L, stay in q1	write 1, move L, stay in q1	write ⊔, move R, go to q_{acc}

6.1 Answer the questions about M.

6.1.1 Give the sequence of configurations of M on the input 011. (3)

6.1.2 Describe in words what M does to any input. (1)

6.2 Explain the difference between a decidable language and a Turing-recognisable language. (3)

6.3 Prove by contradiction that $A_{TM} = \{\langle M, w\rangle \mid M \text{ is a Turing machine that accepts } w\}$ is undecidable. (6)

6.4 Define the classes P and NP. Explain what it means for a problem to be NP-complete, and give ONE example of an NP-complete problem. (5)

6.5 Explain why P = NP would follow if anyone found a polynomial-time algorithm for a single NP-complete problem. (2)

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TOTAL: 100

END OF QUESTION PAPER